File 35: GT Intern Workstream — Page 4: USACE Nationwide Permit 27 & EPD Buffer Variance Compliance

> ***Hydrologic Engineering & Geomorphic Design Manual Target School**: Georgia Institute of Technology (Georgia Tech) School of Civil & Environmental Engineering **Project Area**: Upper Savannah Basin & Chattahoochee River Spawning Headwaters **Supervising Board**: Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)*

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1. Introduction: The Regulatory Hurdles

Even though our stream restorations result in high ecological benefits, any physical work within a stream channel is legally considered a "discharge of dredged or fill material into waters of the United States." Therefore, before Hunter Morris can bring heavy earthworks equipment onto the property, we must secure two critical state and federal regulatory approvals:

1. **USACE Section 404 Nationwide Permit 27 (NWP 27)**: Authorization for aquatic habitat restoration, establishment, and enhancement activities.
2. **Georgia DNR EPD 25-foot State Waters Buffer Variance**: Under the Georgia Erosion and Sedimentation Act, all land-disturbing activities within 25 feet of state waters require a formal variance.

As a Georgia Tech Environmental or Civil Engineering Intern, you will prepare the technical packages, design sheets, and compliance logs to secure these permits.

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2. Preparing the USACE NWP 27 Pre-Construction Notification (PCN)

You will compile and draft the formal **PCN (Form ENG 4345)** package to the USACE Savannah District.

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                    NWP 27 APPLICATION SUBMISSION STEPS

  [ Step 1: Complete ENG 4345 ] ---> [ Step 2: Delineate Jurisdictional Wetlands ]
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  [ Step 4: Verify No-Rise ]    <--- [ Step 3: Attach AutoCAD geomorphic designs ]
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            v
  [ Step 5: Secure GSWCC Level II soil erosion approval ] ---> [ SUBMIT PCN TO USACE ]
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Essential PCN Contents to Compile:

- **Project Description**: Clearly state that the sole purpose of the project is ecological stream restoration using Natural Channel Design (NCD) and native plantings, meeting NWP 27 qualifying criteria.
- **Jurisdictional Wetland Delineation**: Map any local wetlands inside the work corridor according to the *1987 USACE Wetland Delineation Manual*.

- **Impact Matrices**: Calculate the total area of temporary streambed disturbance (in square feet) and length of channel modifications (in linear feet).
- **HEC-RAS No-Rise Statement**: Attach your HEC-RAS floodway modeling output proving that the project causes a 0.00-foot rise in the 100-year flood elevation.
- **Avoidance & Minimization Log**: Describe the construction sequencing steps (e.g., pumping water around the active work reach) to minimize temporary turbiditic sediment releases downstream.

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3. Soil Erosion & Sedimentation Control (GSWCC Level II Standards)

Any land disturbance in Georgia requires a strict **Erosion, Sedimentation, and Pollution Control (ES&PC) Plan** satisfying the standards of the **Georgia Soil and Water Conservation Commission (GSWCC)** "Green Book."

Core ES&PC Design Elements to Draft:

- **Silt Fencing (Sd1-S / Sd1-C)**: Place Type S (Sensitive) or Type C (High-Strength) silt fencing along all construction boundaries to catch sediment runoff.
- **Turbidity Curtains (Tc)**: Specify floating aquatic turbidity curtains downstream of active grading areas to trap suspended clays in the water column.
- **Stream Bypass Channels (Db)**: Detail a temporary pump-around system. Water must be diverted from upstream of the work zone, passed through a flexible pipeline, and discharged downstream through a sediment filter bag (Fb) before returning to the main channel.
- **Immediate Stabilization (Ds3 / Ds4)**: Specify that all disturbed soil banks must be stabilized with coir fiber matting (Type 700) and hydro-seeded with native grass seed mixes within exactly **48 hours** of initial grading.

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4. Georgia DNR EPD 25-Foot Stream Buffer Variance

Under O.C.G.A. § 12-7-6(b)(15), any land disturbance within 25 feet of trout streams requires a state buffer variance.

A. EPD Variance Types

- **Buffer Variance Category 1**: For projects involving minor buffer disturbances that do not modify the stream channel (e.g., planting native buffers).
- **Buffer Variance Category 2**: For projects involving geomorphic modifications, channel re-alignment, or bioengineered instream structures (our typical restoration projects fall under Category 2).

B. Compilation and Review Pipeline

1. **Application Form**: Complete the formal *DNR EPD Stream Buffer Variance Application Form*.
2. **Buffer Mitigation Plan**: Since the project temporarily disturbs the existing buffer to install stable bank slopes, you must detail the **Riparian Planting Plan** (from File 30). Prove that the post-restoration native canopy coverage exceeds the pre-restoration degraded coverage by *\ge 150\%*.

3. **EPD Review Coordinates**: Submit the package to the Georgia DNR EPD Land Protection Branch. The state review takes approximately **60 to 90 days** and includes a 30-day public comment notice period. Monitor the EPD permit portal closely and immediately resolve any comments from state regulators.